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SUPERSEDING
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MILITARY SPECIFICATION

PLASTIC SHEET, FILLED PHENOLIC, UNCURED

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 **Scope.** This specification covers uncured, filled-phenolic, plastic molding sheet used for molding parts for electronic, communications, and mechanical equipment. (See 6.1.) These sheets are not cured when furnished, and therefore, may be formed into the desired shape by molding, at which time the property values specified herein are obtained. (See table I.)

1.2 **Classification.** The uncured molding sheet covered by this specification shall be of one type, PMS.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

L–P–406 — Plastics, Organic: General Specifications, Test Methods.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** Uncured molding sheet shall be composed on an organic or inorganic short-fiber filler impregnated with phenol-formaldehyde resin or binder, and shall be so formulated as to meet the requirements of this specification, when thoroughly cured.

3.1.1 *Uniformity.* Sheets of corresponding thicknesses required for processing shall be uniform in texture, finish, and in the specified properties as determined by the acceptance tests. (See 4.3.)

3.1.2 *Surface defects.* Sheets shall be free from blisters, wrinkles, cracks, and from the inclusion of foreign material.

3.1.3 *Color.* The color of the sheets shall be black.

3.1.4 *Dimensions.* The length, width, and thickness of the sheets shall be as specified. Standard-size sheets from which specimens

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have been cut for tests required by this specification will be accepted unless particular sheet dimensions are specified in the contract or order. (See 6.2.)

3.2 **Property values.** When specimens are tested as specified herein, the values obtained from each set of specimens for a specific property, after each conditioning procedure specified in table I, shall be averaged, and the results so obtained shall meet the required values specified in table I. (See 4.4.2 to 4.4.2.5, incl.)

TABLE I. *Property values of specimens made from molding sheet*

Property to be tested (see table VI)	Conditioning procedure (see 4.4.1.5)	Value required
Water absorption, maximum, percent[1]	See 4.4.2.2	2.0
	See 4.4.2.2	0.40
Insulation resistance, minimum, megohms	C-96/35/90	12
Dielectric breakdown, minimum, kilovolts	D-48/50+D-½ '23	15
Impact strength, minimum foot-pounds per inch of notch:		
Flatwise	E-48/50+C-96/23/50	1.50
Edgewise	E-48/50+C-96/23/50	0.50
Tensile strength, minimum, psi:		
Lengthwise	C-96/23/50	15,000
Crosswise	C-96/23/50	8,500
Flexural strength (flatwise), minimum, psi:		
Cut lengthwise	C-96/23/50	15,000
Cut lengthwise	E-48/50+C-96/23/50	16,500
Cut crosswise	C-96/23/50	14,000
Compressive strength, minimum, psi	C-96/23/50	22,000
	E-48/50+C-96/23/50	25,000
Bonding strength, minimum, pounds	D-48/50+D-½/23	1,000
Heat distortion, minimum, °C	A	150

[1] See table VI for two thicknesses.

3.3 **Marking.** Each sheet shall be legibly marked with the following:

(a) Type.

(b) Specification number.

(c) Manufacturer's name or trademark.

(d) Date of manufacture.

3.4 **Workmanship.** Plastic molding sheet shall be manufactured in a careful and workmanlike manner. It shall be free from any defects which may affect its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Test conditions.** Measurements and tests specified in table II shall be made at room ambient temperature, pressure, and humidity. Unless otherwise specified herein, all other measurements and tests shall be made at 23°±1°C. and 50±2 percent relative humidity. (See table V.)

4.2 **Test equipment and inspection facilities.** Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required acceptance tests. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government.

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4.3 Acceptance tests. Acceptance tests shall consist of groups A, B, and C tests.

4.3.1 *Inspection lot.* An inspection lot shall consist of all sheets manufactured and processed from the same batches of materials under substantially the same conditions and offered for inspection at the same time.

4.3.2 *Group A tests.* Group A tests shall consist of visual and dimensional inspection, as specified in table II. Statistical sampling and inspection shall be in accordance with Standard MIL–STD–105. The acceptable quality levels (AQL) shall be as specified in table II. Major and minor defects shall be as defined in Standard MIL–STD–105.

TABLE II. *Group A acceptance tests*

Test	Requirement paragraph	Test paragraph	AQL (percent defective)	
			Major	Minor
Visual and dimensional inspection:	....	4.4.2.1	1.0	4.0
Material	3.1	....		
Uniformity	3.1.1	....		
Surface defects	3.1.2	....		
Color	3.1.3	....		
Dimensions	3.1.4	....		
Marking	3.3			
Workmanship	3.4			

4.3.3 *Group B tests.* Group B tests shall consist of those specified in table III. They shall be performed on specimens cut from sample sheets that have been subjected to and have passed the group A tests, unless the Government considers it more practical to select separate sample sheets from the lot for group B tests. All sample sheets shall be selected without regard to their quality.

TABLE III. *Group B acceptance tests*

Test	Conditioning Procedure (see 4.4.1.5)	Requirement table	Test table
Water absorption	See 4.4.2.2	Table I	Table VI
Dielectric breakdown	D–48/50+D–½/23		
Flexural strength (flatwise): Cut crosswise	C–96/23/50		

4.3.3.1 *Sampling procedure.* The sampling procedure shall be in accordance with the appendix to Standard MIL–STD–105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. The AQL shall be 6.5 (percent defective) and the inspection level shall be L7 for normal inspection and L5 for reduced inspection. The total number of defectives for the tests listed in table III shall be compared with the acceptance number for that sample size. If the acceptance number is exceeded, the lot shall be rejected. When the

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inspection lot does not contain ⅛- or ½-inch thick sheets, or both, the required thickness shall be prepared as follows: 6-inch strips shall be cut from each of the shorter edges of uncured sheets. These strips shall then be cut into 6-inch squares, and sufficient of the squares from each sheet shall be laminated together by methods representative of sound production practice to produce squares of the requisite thickness to represent the sheets.

4.3.3.2 *Action in case of lot rejection.* If an inspection lot is rejected due to failure to pass group B tests, the manufacturer may rework the inspection lot or screen out defectives and resubmit it for Government inspection. Resubmitted inspection lots shall be kept separate from new inspection lots. The resubmitted lots shall be inspected, using tightened inspection.

4.3.4 *Group C tests.* Group C tests shall consist of those specified in table IV. They shall be performed on specimens cut from sample sheets that have been subjected to and have passed the group A tests, unless the Government considers it more practical to select separate sample sheets from the lot for group C tests.

TABLE IV. *Group C acceptance tests*

Test	Conditioning procedure (See 4.4.1.5)	Requirement table	Test table
Insulation resistance	C-96/35/90	Table I	Table VI
Impact strength:			
Flatwise	E-48/50+C-96/23/50		
Edgewise	E-48/50+C-96/23/50		
Tensile strength:			
Lengthwise	C-96/23/50		
Crosswise	C-96/23/50		
Flexural strength (flatwise):			
Cut lengthwise	C-96/23/50		
Cut lengthwise	E-48/50+C-96/23/50		
Compressive strength	C-96/23/50		
	E-48/50+C-96/23/50		
Bonding strength	D-48/50+D-½/23		
Heat distortion	A		

4.3.4.1 *Sampling procedure.* A sample sheet shall be selected without regard to quality from the first inspection lot, and thereafter from those inspection lots in which every 2,000th sheet occurs, or once every 2 months, whichever is less frequent, and specimens for the group C tests shall be cut therefrom. If the average value for any group C test falls outside the specification limits, the inspection lot shall be rejected. When the inspection lot does not contain ⅛- or ½-inch thick sheets, or both, the required thickness shall be prepared as follows: 6-inch strips shall be cut from each of the shorter edges of uncured sheets. These strips shall then be cut into 6-inch squares, and sufficient of the squares from each sheet shall be laminated together by methods representative of sound production practice to produce squares of the requisite thickness to represent the sheets.

4.3.4.2 *Noncompliance.* If a sample fails group C tests, the manufacturer shall take corrective action on the processes and all units of product which were manufactured

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with the same conditions, materials, processes, etc, and are considered subject to the same test failure. Government inspection may be discontinued, at the option of the procuring agency, technical bureau, or service, until a sample produced by the corrected process has been subjected to and has passed the group C tests failed by the preceding sample. Units of product subsequently submitted for Government inspection shall incorporate the corrective action, unless this requirement is waived temporarily by the procuring agency, technical bureau, or service.

4.3.5 *Defective sheets.* Sheets which have been found defective on acceptance testing shall not be furnished under contract or order even though the lot is acceptable.

4.3.6 *Results of test.* When specified (see 6.2), a certified copy of the results of the acceptance tests shall be furnished the Government.

4.4 Test procedures.

4.4.1 *Specimens.*

4.4.1.1 *Number.* The number of specimens to be tested per sample sheet shall be as specified in table VI. The results of the tests on the specimens taken from each sample sheet shall be averaged and that average shall determine whether or not the individual sample sheet shall be counted as defective.

4.4.1.2 *Form and dimensions.* The form and dimensions of specimens shall be as specified in the applicable method of Specification L-P-406 and as modified in table VI.

4.4.1.3 *Preparation.* Specimens shall be die-cut or machined from ⅛- and ½-inch thick compression-molded sheets. (See 4.3.3.1, 4.3.4.1, and table VI.) The direction of grain, lengthwise and crosswise, shall be clearly defined.

4.4.1.4 *Tolerances.* Unless otherwise specified herein, standard commercial tolerances shall apply.

4.4.1.5 *Conditioning.* Specimens shall be conditioned before test, as specified in table I. (See 4.4.1.5.1 to 4.4.1.5.3, incl.)

4.4.1.5.1 *Designation.* The type of conditioning required shall be designated as follows:

Condition A—As received; no special conditioning.

Condition C—Humidity conditioning.

Condition D—Imersion conditioning in distilled water.

Condition E—Temperature conditioning.

4.4.1.5.2 *Procedures.* The conditioning procedures required, with the exception of condition A, shall be indicated by the following combination of symbols:

(a) A capital letter indicating the type of conditioning.

(b) A number indicating in hours the duration of the conditioning.

(c) A number indicating in degrees centigrade the conditioning temperature.

(d) A number indicating relative humidity, whenever relative humidity is controlled.

The numbers shall be separated from each other by a slant mark, and from the capital letter by a dash. A sequence of conditions shall be denoted by the use of a plus (+) sign between successive conditions.

4.4.1.5.3 *Tolerances.* Conditioning tolerances and additional testing information shall be as specified in table V.

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TABLE V. *Conditioning tolerances and testing conditions*

Procedure designation	Conditioning tolerances			Testing conditions	
	Time, hours	Temperature, °C.	Humidity, percent	Temperature, humidity, and tolerances	Remarks
C–96/23/50	−0, +2	±5	±5	23°±5°C., 50±5 percent RH.	....
C–96/35/90	−0, +2 (chamber)	±2	±2	35°±2°C., 90±2 percent RH.	Start test immediately.[1]
D–48/50+D–½/23	−0, +2 (hot water), −0, +2 (cool water).	±5 ±5	....	23°±5°C., 50±5 percent RH.	Start test within 2 minutes after removing specimens from final conditioning bath.[1]
E–48/50+C–96/23/50	−0, +2 (oven), −0, +2	±5 ±5	.... ±5	23°±5°C., 50±5 percent RH.	....

[1] After immersion or 90-percent RH conditioning, surface water shall be removed by wiping the specimen with a damp cloth, followed by wiping with a dry cloth.

4.4.1.6 *Measurement of specimens.* Specimens shall be examined, and dimensions required to be recorded shall be measured to the nearest 0.001 inch by using a suitable micrometer or vernier caliper. All other dimensions shall be measured to the nearest 1/32 inch. Measurements shall be made before conditioning in each case and after completion of the test whenever required.

TABLE VI. *Test methods and specimens*

Test (see table I)	Spec. L–P–406 Method No.	Modified by paragraph	Specimens	
			Number per sample sheet	Form and dimensions
Water absorption	7031	4.4.2.2	3	3×1×⅛ inches
			3	3×1×½ inches
Insulation resistance	4041	4.4.2.3	3	(See fig. 1)
Dielectric breakdown	4031	4.4.2.4	5	(See fig. 1)
Impact strength:	1071	....		(See Spec L–P–406)
Flatwise			5	
Edgewise			5	
Tensile strength:	1011	....		(See Spec L–P–406)
Lengthwise			5	
Crosswise			5	
Flexural strength (flatwise):	1031.1	....		(See Spec L–P–406)
Cut lengthwise			*10	
Cut crosswise			3	
Compressive strength	1021.1	....	*10	1×½×½ inch
Bonding strength	1111	....	5	(See Spec L–P–406)
Heat distortion	2011.1	4.4.2.5	3	(See Spec L–P–406)

* Five specimens for each conditioning procedure. (See table I.)

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4.4.2 *Test methods.* The test methods and the number, form, and dimensions of the specimens to be tested shall be as specified in table VI, and as modified therein.

4.4.2.1 *Visual and dimensional inspection.* Sheets shall be inspected to verify that the material, uniformity, surface defects, color, physical dimensions, marking, and workmanship are in accordance with applicable requirements. (See 3.1 to 3.1.4, incl, 3.3, and 3.4.)

4.4.2.2 *Water absorption.* Specimens shall be tested in accordance with method 7031 of Specification L-P-406. The following exceptions shall apply:

(a) Oven conditioning at 105°C. for 1 hour.

(b) Cooled in a desiccator for at least 2 hours.

(c) Soluble matter loss—Not applicable.

4.4.2.3 *Insulation resistance.* Electrodes shall be American Standard No. 3 tapered pins. (See fig. 1.)

4.4.2.4 *Dielectric breakdown.* The test shall be made under oil at a frequency not exceeding 100 cycles per second. Electrodes shall be American Standard No. 3 tapered pins. (See fig. 1.)

4.4.2.5 *Heat distortion.* Specimens shall be placed directly in an oil bath and not in an air bath surrounded by an oil bath.

4.5 **Test reports.** Test reports shall be made as specified in Specification L-P-406; in addition, a summary sheet of test results shall be furnished.

5. PREPARATION FOR DELIVERY

5.1 Not applicable to this specification.

6. NOTES

6.1 **Intended use.** Type PMS plastic molding sheet is intended for general-purpose use. It may be used alone or in conjunction with molding material. Typical applications of this type molding sheet are: telegraphkey bases, instrument dials, motor housings, switch-controller parts, instrument panels, etc.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Length, width, and thickness required. (See 3.1.4.)

(c) Whether a certified copy of the results of the acceptance tests be furnished the Government. (See 4.3.6.)

6.3 *Curing. Final curing, by processors, of* the uncured plastic molding sheet should be accomplished by using the temperature, time, and pressure recommended by the manufacturer of the material.

Patent notice.—When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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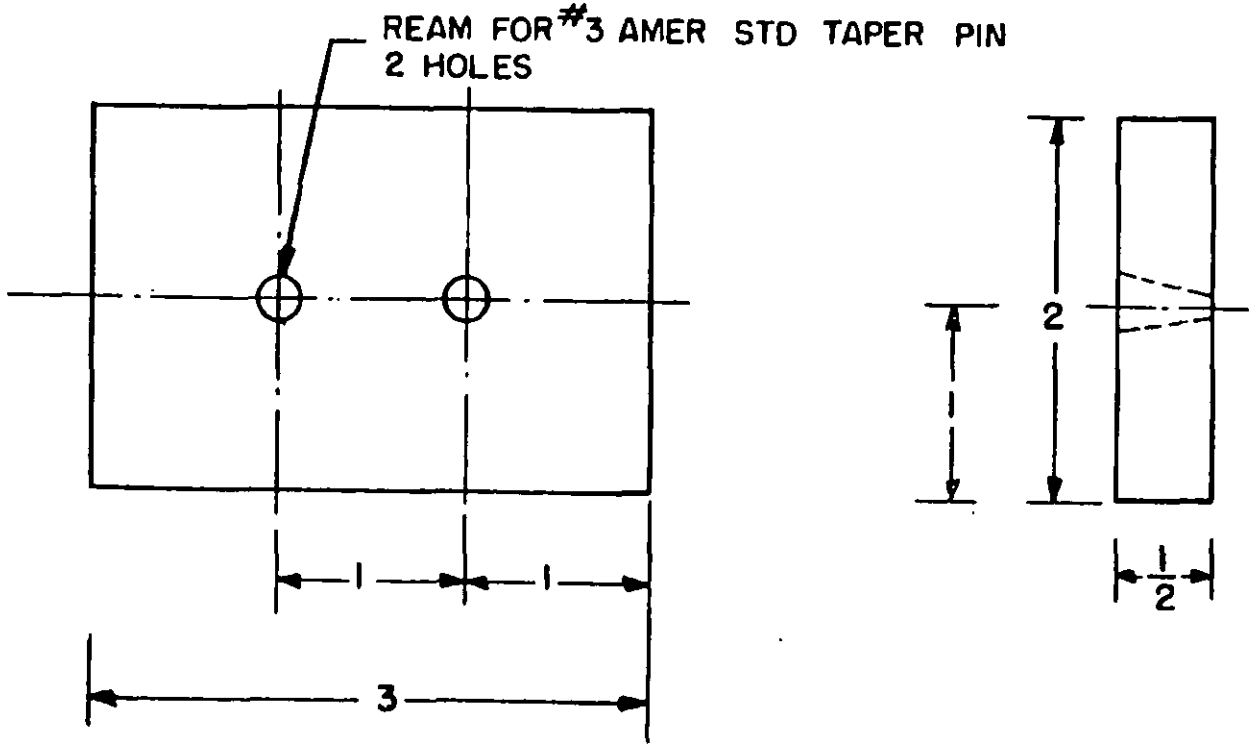


FIGURE 1.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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